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Carmen, Cagayan de Oro City

CHERRY COSMETICS

DEFENSE STUDY GUIDE

Desktop-Based Ordering System for Cherry Cosmetics Shop

Prepared for project review and oral defense

Course: ITE-298 Information Management (Database Systems 1 and 2)

PROJECT TEAM

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Simple explanations • Key details • Possible questions and answers

How to Use This Guide

Use this guide to understand the project in your own words. Do not memorize every sentence exactly. Learn the main ideas, then explain them naturally during the presentation.

Study tip: For most defense questions, answer in three parts: what the feature is, how Cherry Cosmetics uses it, and why it is useful.

1. Project at a Glance

What is the project?

The Desktop-Based Ordering System for Cherry Cosmetics Shop is a Java desktop application that helps customers browse cosmetic products, place orders, and check order progress. It helps the administrator manage products, customer accounts, orders, payment records, stock, and sales reports. SQLite stores the information.

What problem does it solve?

- Orders and questions arrive through Messenger conversations, so older messages can be hard to find.
- Product, customer, order, and payment details may need to be written down manually.
- The customer may not have a dedicated place to check order status.
- Stock quantities can become difficult to track when updates are manual.

What is the proposed solution?

The system keeps ordering and store records in one desktop application and one SQLite database. Customers use the customer screens, while the administrator uses the management screens.

Main objective

To make product browsing, ordering, payment recording, inventory tracking, and record management more organized and easier to handle.

2. Users and Main Features

User / Module	What it does
Customer account	Register, log in, view products, add items to the cart, place orders, view order history and status.
Product management	Admin adds, views, updates, and archives products; manages price, discount, category, image, description, and stock.
Customer management	Admin searches, sorts, views, and archives customer accounts.
Cart and checkout	Customer chooses quantities, reviews the cart, provides location/contact details, selects a payment method, and submits an order.
Order management	Admin views orders and updates their statuses.
Payment records	Stores payment method, status, reference, and payment date; it does not charge a bank account or process a real online payment.

User / Module	What it does
Inventory	Tracks available stock; stock can be restored for eligible cancellations and approved returns.
Sales reports and export	Shows weekly, monthly, and yearly reports, with filtering. The displayed report can be exported to CSV.
Receipt	Can create a PNG receipt for eligible orders.

3. Technologies Used

Technology	Simple explanation	Why used
Java	Programming language used for the application logic.	Handles login, products, orders, stock, payment records, and reports.
Java Swing / Java GUI	Tools for the desktop screens, forms, buttons, and tables.	Creates the interface customers and admins use.
NetBeans IDE	Program for writing and running the Java project.	Development environment for the team.
SQLite	A file-based relational database.	Stores the project records in data/cherry_cosmetics.db.
SQLite JDBC driver	Connects Java to SQLite.	Allows the Java program to read and change database records.
DB Browser for SQLite	A graphical tool for opening and inspecting an SQLite file.	Helps verify saved records and demonstrate archive behavior.

Remember: The main application is desktop-based, not a website. The Java application uses JDBC to connect to SQLite.

4. Database in Simple Words

A database is an organized place to store information. A table stores one kind of record. A column describes a detail, and a row is one saved record.

Table	What it stores	Example fields
USERS	Admin and customer accounts.	id, username, email, password, role, is_archived
CATEGORIES	Product category names.	id, name, is_archived
PRODUCTS	Cosmetic product details.	id, name, category_id, price, stock, discount, image, is_archived
ORDERS	Main details for each customer order.	id, customer_id, order_date, status, total
ORDER_ITEMS	Products and quantities inside an order.	id, order_id, product_id, qty, unit_price
PAYMENTS	Payment record associated with an order.	id, order_id, method, status, reference, payment_date

Key database terms

Term	Simple meaning	Cherry Cosmetics example
Primary Key (PK)	A unique ID for each row.	users.id identifies one account.
Foreign Key (FK)	A column that refers to a row in another table.	orders.customer_id refers to users.id.
UNIQUE	Prevents duplicate values for the specified column or combination.	username and email are unique; payments.order_id is unique.
NOT NULL	Requires a value to be present.	A product must have a name.
CHECK	Requires a value to pass a rule.	price >= 0; qty > 0.
Index	A structure that can help the database find rows faster.	Indexes are defined for product category, order customer, and order item links.
DDL	SQL commands that define database structures.	CREATE TABLE and CREATE INDEX.
Junction table	Connects two tables in a many-to-many relationship.	ORDER_ITEMS connects ORDERS and PRODUCTS.

How the tables are connected

- USERS → ORDERS: one customer can place many orders; each order points to one user.
- CATEGORIES → PRODUCTS: one category can contain many products; each product references one category.
- ORDERS → ORDER_ITEMS: one order can contain multiple order-item rows.
- PRODUCTS → ORDER_ITEMS: the same product can appear in multiple order-item rows.
- ORDERS and PRODUCTS: this many-to-many relationship is resolved by ORDER_ITEMS.
- ORDERS → PAYMENTS: an order can have zero or one payment record because payments.order_id is unique.

Why is ORDER_ITEMS needed?

One order can contain several products, and a product can appear in different orders. ORDER_ITEMS saves one row for each product included in an order, including its quantity and unit price. This avoids putting a list of products into one field.

5. Normalization: 1NF, 2NF, and 3NF

Rule	Easy explanation	Project example
1NF	Each field holds one value, not a list of values.	Each ordered product is stored in its own ORDER_ITEMS row.
2NF	Each non-key field should depend on the whole primary key.	Quantity and unit_price describe one order item, while general order details are in ORDERS.
3NF	Non-key fields should depend on the key, not on another non-key field. Keep separate kinds of information in suitable tables.	Category names are in CATEGORIES; product details are in PRODUCTS; payment details are in PAYMENTS.

Simple answer: Normalization organizes data into related tables to reduce unnecessary duplication and update problems.

6. How an Order Is Processed

1. The customer logs in and browses the available products.
2. The customer chooses products and quantities and adds them to the cart.
3. The system calculates the order total and checks available stock.
4. The customer enters required location and contact details and chooses Cash on Delivery or GCash.
5. If GCash is selected, the customer provides the required reference number.
6. The system saves the order, order items, and payment record in SQLite.
7. The application updates stock quantities separately.
8. The order begins with Pending status and can be viewed in My Orders.

Be accurate: The documentation says stock changes are saved separately after the order, order items, and payment records are saved. Do not claim all these operations are one single transaction unless your current code is changed and tested.

7. Archive / Soft Delete

Archiving is different from permanently deleting a record. In a soft archive, the application changes the is_archived value from 0 to 1. The record remains in the database, but it should no longer appear in the normal active list.

Value	Meaning
is_archived = 0	Active record
is_archived = 1	Archived record

In the current interface, Archive is available for customer accounts and products. The schema also has an archive field for categories, but do not claim that the category screen has a working archive button unless you have verified it.

Presentation demonstration: open the exact same data/cherry_cosmetics.db file in DB Browser for SQLite, find a product, archive it in the application, refresh Browse Data, and show that the same product ID still exists with is_archived = 1.

8. Reports, CSV, and Receipts

- The admin report screen supports weekly, monthly, and yearly reporting, along with search and status filters.
- CSV is a text-based file format that stores rows and columns and can be opened in spreadsheet tools. The application exports the report currently displayed in the table.
- The system can generate an eligible receipt as a PNG image.
- A database VIEW is not required for CSV export. The project currently documents six tables and zero database views.

9. Scope and Limitations

Included

- Customer registration and login.
- Product browsing, search, category filtering, and cart.
- Order placement and order-status tracking.
- Admin management of products, customers, and orders.
- Recording payment methods, statuses, and references.
- Stock management and basic sales reports.
- CSV report export and eligible PNG receipts.

Limitations

- The application runs on desktop and is not a mobile or browser-based ordering website.
- Payment details are recorded, but no payment gateway processes actual online payments.
- Stock accuracy depends on application logic and the administrator keeping product information up to date.
- Reports are basic, not advanced business analytics.
- The current code stores passwords as plain text; this is a security limitation that should be improved with password hashing.

10. Methodology and Work Plan

The documents describe development in small stages with Scrum practices. The team divides work into smaller tasks, develops features, tests them, reviews progress, and improves the system based on results or feedback.

Scrum term	Simple explanation
Product backlog	List of features and tasks the team wants to complete.
Sprint	A short development period focused on a set of tasks.
Sprint planning	The team selects and plans tasks for the next sprint.
Team check-in / Daily Scrum	Members share progress and blockers.
Sprint review	The team demonstrates completed work and gets feedback.
Sprint retrospective	The team discusses how to improve its way of working.

The project plan describes a seven-week schedule: requirements and planning, interface/database design, customer/product development, ordering/payment development, admin/inventory functions, testing, then final improvements and documentation. The estimated budget is PHP 2,800.

11. Security and Reliability

- Prepared statements are used for many SQL operations to reduce SQL injection risk when handling user-provided values.
- The database has constraints such as unique usernames/emails and checks on prices, discounts, stock, and quantities.
- Passwords are currently stored as plain text, which is not safe for production. A future version should use secure password hashing.
- Keep backup copies of the project folder and database file, especially before a live demonstration.

12. Key Facts to Memorize

Question	Answer to remember
Project type	Desktop-based ordering system.
Client	Cherry Cosmetics Shop.
Programming language / GUI	Java with Java Swing.
IDE	NetBeans.
Database	SQLite.
Database file	data/cherry_cosmetics.db.
Tables	Six: USERS, CATEGORIES, PRODUCTS, ORDERS, ORDER_ITEMS, PAYMENTS.
Indexes in the documented schema	Four.
Reporting views	Zero.
Junction table	ORDER_ITEMS.
Payment gateway	None; payment details are recorded only.
Report export	CSV.
Receipt output	PNG.
Estimated project budget	PHP 2,800.
Planned schedule	Seven weeks.

13. Possible Defense Questions and Simple Answers

Practice answering in your own words. These are suggested answers based on the project documents. Be ready to explain or demonstrate each feature in the running application.

1. What is your project about?

Answer: It is a desktop ordering system for Cherry Cosmetics Shop. It helps customers order products and lets the admin manage products, customers, orders, payments, stock, and sales reports.

2. Why did you choose this project?

Answer: The shop mainly receives orders through Messenger. When many messages arrive, old order details can be difficult to find. The system puts important records in one organized place.

3. What problem does your system solve?

Answer: It helps reduce scattered order records, makes order tracking easier, and gives the admin a better way to manage products and stock.

4. Who are the users?

Answer: The two main user types are customers and the administrator.

5. What can customers do?

Answer: They can register, log in, browse products, use the cart, place orders, and check their order history and status.

6. What can the admin do?

Answer: The admin can manage products and customer accounts, process orders, update stock, check payment records, and view sales reports.

7. Why is it desktop-based?

Answer: The application was built with Java Swing for a desktop computer. It is not a mobile app or a browser-based website.

8. What technologies did you use?

Answer: Java and Java Swing for the application, NetBeans for development, SQLite for storage, and JDBC to connect Java to SQLite.

9. What is SQLite?

Answer: SQLite is a relational database stored in a file. Our project data is stored in cherry_cosmetics.db.

10. What is JDBC?

Answer: JDBC is the Java technology that lets the application connect to the database and run SQL commands.

11. How many tables are in your database?

Answer: There are six main tables: USERS, CATEGORIES, PRODUCTS, ORDERS, ORDER_ITEMS, and PAYMENTS.

12. What is a primary key?

Answer: A primary key is a unique identifier for each row, such as id in the USERS table.

13. What is a foreign key?

Answer: A foreign key connects a record to a related record in another table. For example, orders.customer_id references users.id.

14. Why do you need ORDER_ITEMS?

Answer: An order can contain multiple products, and the same product can appear in many orders. ORDER_ITEMS connects ORDERS and PRODUCTS and stores quantity and unit price.

15. What is a many-to-many relationship in your system?

Answer: ORDERS and PRODUCTS have a many-to-many relationship, resolved through ORDER_ITEMS.

16. Why is payments.order_id unique?

Answer: It means each order can have at most one payment record in this schema.

17. What is normalization?

Answer: Normalization organizes information into related tables to reduce repeated data and prevent update problems.

18. Explain 1NF in simple terms.

Answer: Each field stores one value. Product lists for an order are stored as separate ORDER_ITEMS rows instead of a list in one field.

19. Explain 2NF in simple terms.

Answer: Each non-key field should depend on the whole key for its record. Order-item quantity and unit price belong to a specific order item.

20. Explain 3NF in simple terms.

Answer: Information is stored in suitable related tables so non-key details do not depend on other non-key details. For example, category names are stored in CATEGORIES.

21. What happens when a customer places an order?

Answer: The customer selects products and quantities, enters checkout details, chooses a payment method, and confirms. The system saves the order, order items, and payment record, then updates stock.

22. Does the system process actual GCash or online payments?

Answer: No. It records the chosen payment method, status, and reference. It does not connect to a payment gateway to transfer or collect money.

23. What is soft deletion or archiving?

Answer: It means the record is marked as archived instead of being permanently deleted. The is_archived value changes from 0 to 1.

24. How can you prove the archived record still exists?

Answer: I can archive a product in the app, open the same database file in DB Browser, refresh the table, and show that its ID still exists with is_archived set to 1.

25. Do all tables have an Archive button?

Answer: No. The current interface supports archiving products and customer accounts. Not every table needs its own archive button, especially transaction-detail tables.

26. What is a database index?

Answer: An index can help the database locate records more efficiently for common lookups. Our documented schema has four indexes.

27. What is a database view?

Answer: A view is a saved SQL query that acts like a virtual table. Our current database does not define reporting views; reports use queries directly.

28. What is CSV export used for?

Answer: It saves the currently displayed sales report as a CSV file that can be opened in spreadsheet software.

29. What is the receipt file format?

Answer: An eligible receipt can be generated as a PNG image.

30. What are the project limitations?

Answer: It is desktop-only, does not process real online payments, has basic reporting, and depends on accurate stock maintenance. Passwords also need stronger protection.

31. What methodology do you use?

Answer: The project documents describe iterative development with Agile/Scrum practices: plan small tasks, develop, test, review, and improve.

32. What is a sprint?

Answer: A sprint is a short development period in which the team works on selected tasks or features.

33. How long is the planned development schedule?

Answer: The project schedule is seven weeks.

34. What is the estimated project budget?

Answer: The estimated total is PHP 2,800.

35. How do you reduce SQL injection risk?

Answer: Many database operations use prepared statements so user-entered values are handled as parameters rather than directly joined into SQL text.

36. Is storing passwords as plain text secure?

Answer: No. It is a known limitation of the current implementation. A stronger version should store password hashes rather than plain-text passwords.

37. What happens if the project database is lost?

Answer: The system may lose its saved records, so the team should keep backup copies of the database and project folder.

38. How did you test the database?

Answer: We can create or update records through the application, then check the same SQLite database in DB Browser to confirm that the data was saved.

39. Why not use a database view for reports?

Answer: The project can generate reports with SQL queries directly. A view is optional and is not required by the stated schema.

40. What makes your system useful to the client?

Answer: It keeps product, customer, order, payment, and stock information organized in one system and makes order status easier to track.

14. Important Items to Verify Before the Presentation

These are not new features to add. They are small consistency checks so the proposal, technical manual, database document, and running application tell the same story.

Item	What to check
Database name	Use the filename data/cherry_cosmetics.db consistently. Avoid using cherry_cosmetic_db as if it were the actual filename.
Development method	Some document sections say Iterative SDLC, while others say Agile SDLC with Scrum. Agree on one consistent explanation. A safe summary is iterative development in small stages using Scrum practices, if that matches what the team actually did.
Default admin password	The documents show admin / Password123, but the code or seeded database may use a different password. Test the exact final build and write only the working credentials in the manual.
Archive demonstration	Verify the product/customer record in the same database file the running app uses. Do not accidentally inspect a different copy in the XAMPP htdocs folder.
PHP requirement, if the instructor requires it	The desktop Java app connects through JDBC. A separate PHP API file is not proof that the Java app uses PHP; demonstrate the PHP connection separately if required by the rubric.
Checklist accuracy	Do not claim that views exist or that all testing/debugging cleanup is complete unless the final files actually confirm it.
Feature claims	Only say a feature works if you can show it in the current running system.

15. A Short Practice Introduction

Suggested opening: Good day. Our project is the Desktop-Based Ordering System for Cherry Cosmetics Shop. We developed it to help organize product information, customer orders, payment records, stock, and sales reports. The application uses Java Swing and connects to an SQLite database through JDBC. Customers can browse products and place orders, while the administrator can manage products, customers, orders, and reports. The goal is to make the shop's ordering and record-keeping process easier to manage.

16. Last-Minute Defense Checklist

- Know the project problem, purpose, users, and main features.
- Memorize the six table names and what each table stores.
- Be able to explain PK, FK, ORDER_ITEMS, and the 1NF–3NF ideas in simple words.
- Run the final system and confirm the login credentials before presenting.
- Use DB Browser to demonstrate that archiving keeps the record in the database.
- Demonstrate a basic create/update/read action and show that the data is saved.
- Know what the system does not do: it does not process actual online payments.
- Keep a backup copy of the project and database before the presentation.
- Answer honestly if a feature is limited or still needs improvement.